

TOURNAMENT OF TOWNS

31st Tournament • Academic Year 2009–2010

Fall (Autumn) Round • 25 October 2009 • Junior (Grades 8–9) • Advanced (A-Level)

Score is based on the three problems with the best results; points for sub-parts of a problem are summed.

Problem 1. [4 points]

In 10 identical jugs, milk was poured — not necessarily evenly, but each was filled no more than 10%. In one operation, you can choose a jug and pour any part of it evenly into the other jugs. Prove that in no more than 10 such operations, it is possible to achieve equal amounts of milk in all jugs. [Diagram]

(E.N. Gorinov)

Problem 2. [6 points]

Misha has 1000 identical cubes, each with one pair of opposite faces white, the second pair blue, and the third pair red. He assembled a large $10 \times 10 \times 10$ cube from them by applying the cubes to each other with same-colored faces. Prove that the large cube has a monochromatic face.

(M.V. Murashkin)

Problem 3. [6 points]

Find all natural numbers a and b such that $(a+b^2)(b+a^2)$ is an integer power of two.

(V.V. Proizvolov)

Problem 4. [6 points]

On the sides BC and CD of rhombus $ABCD$, points P and Q were chosen such that $BP = CQ$. Prove that the intersection point of the medians of triangle APQ lies on the diagonal BD of the rhombus. [Diagram]

(V.V. Proizvolov)

Problem 5. [9 points]

From weights of 1 g, 2 g, ..., N g, it is required to select several (more than one) with a total weight equal to the average weight of the remaining weights. Prove that: a) this can be done if $N + 1$ is a square of an integer; b) if this can be done, then $N + 1$ is a square of an integer.

(A.V. Shapovalov)

Problem 6. [10 points]

2009 identical squares were placed on a grid plane, with their sides along the grid lines. Then, all cells that are covered by an odd number of squares were marked. Prove that the number of marked cells is not less than the number of cells in one square.

(I. Pak, Yu. Rabinovich)

Problem 7. [14 points]

Olya and Maxim paid for a trip to an archipelago of 2009 islands, where some islands are connected by two-way catamaran routes. They travel, playing a game. First, Olya chooses the island they arrive at. Then they travel together on catamarans, taking turns choosing an island they have not been to before (the first choice is made by Maxim). The person who cannot choose an island loses. Prove that with any scheme of routes, Olya can win, no matter how Maxim plays.

(Folklore, proposed by A.V. Shapovalov)
